Midterm 2

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1. Consider the IBVP:

$$u_t - u_{xx} = f(t)\sin(3x),\ u(0,t) = u(\pi,t) = 0,\ u(x,0) = x - 1,$$

where $0 < x < \pi$, $t > 0$, $f(t) = 1$ for $0 \leq t \leq 1$, and $f(t) = 0$ for $t > 1$.

Proof (Hernandez). We obtain eigenvalues $\lambda_n = n$ and eigenfunctions $y_n = \sin(nx)$ and:

$$u(x,t) = \sum_{n=1}^{\infty} b_n(t)\sin(nx), F(x,t) = \sum_{n=1}^{\infty} f_n(t)\sin(nx), x - 1 = \sum_{n=1}^{\infty} \beta_n \sin(nx).$$

where $f_1(t) = 1$ on $0 \leq t \leq 1$ and $f_n(t) = 0$ otherwise. Substitution yields:

$$\sum_{n=1}^{\infty} \dot{b_n}(t)\sin(nx) - \sum_{n=1}^{\infty} -n^2 b_n \sin(nx) = \sum_{n=1}^{\infty} f_n(t)\sin(nx)$$
$$\dot{b_n}(t) + n^2 b_n = f_n(t),$$

and applying the initial condition $u(x,0) = \sum_{n=1}^{\infty} b_n(0)\sin(nx) = x - 1$ implies $b_n(0) = \beta_n$, so

$$b_n(t) = \beta_n e^{-n^2 t} + \int_0^t e^{-n^2(t-\tau)} f_n(\tau)\ d\tau = \beta_n e^{-n^2 t}$$
$$b_1(t) = \beta_1 e^{-1^2 t} + \int_0^t e^{-1^2(t-\tau)} f_1(\tau)\ d\tau = \beta_1 e^{-t} + 1 - e^{-t} \text{ if } 0 \leq t \leq 1$$
$$b_1(t) = \beta_1 e^{-1t} + \int_0^t e^{-1^2(t-\tau)} f_1(\tau)\ d\tau = \beta_1 e^{-t} + (e-1)e^{-t} \text{ if } t > 1,$$

Thus, we obtain:

$$u(x,t) = \sum_{n=1}^{\infty} b_n(t)\sin(nx),$$

where the b_n are given explicitly in terms of the Fourier coefficients β_n. □

2. Consider the IBVP:

$$u_{tt} = u_{xx},\ 0 < x < 1, 0 < t < 1;\ u_x(0,t) = 1, u_x(1,t) = 0,\ u(x,0) = 1,\ u_t(x,0) = 0.$$

Proof (Hernandez). Suppose $u(x,t) = y(x)g(t)$. Then, we obtain the following ODEs:

$$g''(t) = \lambda_n^2 g(t) y_n''(x) + \lambda_n^2 = 0, \qquad y'(0) = y'(1) = 0,$$

whose general solutions are $g(t) = e^{-\lambda_n^2 t}$ and $y_n(x) = A\sin(\lambda_n x) + B\cos(\lambda_n x)$. Applying the boundary condition $y'(0) = \lambda_n A\cos(\lambda_n \cdot 0) - \lambda_n B\sin(\lambda_n \cdot 0) = \lambda_n A \cdot 1 - \lambda_n B \cdot 1 = 0$ implies $A = 0$. Applying the boundary condition $y'(1) = -\lambda_n \cos(\lambda_n \cdot 1) = 0$ yields:

$$\lambda_n = \arccos\left(\cos(\lambda_n)\right) = \arccos\left(-\frac{0}{\lambda_n}\right) = \pi\left(n - \frac{1}{2}\right).$$

Thus, we obtain:

$$u(x,t) = \sum_{n=1}^{\infty} \cos(\lambda_n x) e^{-\lambda_n^2 t}.$$

□

3. Consider the IBVP:

$$u_{tt} = \Delta u, 0 < r < 1, t > 0;\ u|_{r=1} = 0, u|_{t=0} = 0, u_t|_{t=0} = r.$$

Proof (Hernandez). The first condition $u|_{r=1}(0)$ states the boundary of the disk is held at constant density. The second condition states that the initial temperature is constant. The third condition states that the rate of change of density is linear with respect to the radius. Thus, this is a spherical wave equation, so the general solution is given by:

$$u(r,t) = \frac{1}{r}\left(F(r-ct) + G(r+ct)\right)$$

The first condition requires $F(r-ct) = -G(r+ct)$. The second condition requires $F(r) = -G(r)0$. The third condition requires $G(r+ct) = \frac{r^2}{2c}$ and thus $F(r-ct) = \frac{-r^2}{2c}$. □

4. Consider the wave equation in a ring:

$$u_{tt} = c^2 u_{xx},\ x \in (0,l), t > 0; u(0,t) = u(l,t)\ u_x(0,t) = u_x(l,t), \tag{1}$$

$$u(x,0) = u_0,\ u_t(x,0) = 0. \tag{2}$$

Proof (Hernandez). Consider the following superposition of odd periodic extensions of the constant initial condition:

$$u(x,t) = \frac{1}{2}\left(F(x-ct) + F(x+ct)\right) = \frac{1}{2}\left(u_0 + u_0\right) = \frac{2u_0}{2} = u_0, \qquad \text{where } F(\xi) = u_0.$$

In addition to $u(x,0) = u_0$, the solution also satisfies $u_{tt} = c^2 u_{xx} = 0$, $u(0,t) = u(l,t) = u_0$, $u_x(0,t) = u_x(l,t) = 0$, $u_x(x,0) = 0$ because $u(x,t)$ is a constant whose derivatives are thus zero.

The first row of boundary conditions implies the density is held fixed at its endpoints, and the second row of initial conditions implies the string starts with constant and unchanging density along the string. Therefore, the solution corresponds to a string with constant density u across space and time. □